Uncertainty Quantification in Deep Learning for Radiology Triage

(2024)

Problem

Flagging low-confidence chest CT predictions for radiologist review in a triage setting.

Method

Monte Carlo dropout and deep ensembles used to estimate predictive uncertainty.

Dataset

RSNA pneumonia detection dataset.

Evaluation

Calibration error and triage efficiency (cases correctly deferred to human review).

Results

Uncertainty-based triage reduced radiologist workload by 22% while deferring 95% of true errors for manual review.

Limitations

No prospective clinical trial has evaluated the real-world workflow impact of this triage approach.